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Roll No

EX-702 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Electrical Drives
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe static Kramer drive for speed control of induction motor.
b) Describe static Scherbius drive for speed control of induction motor.
2. A separately excited dc motor takes 50A at a speed of 800 rpm. It has armature resistance of $0.4\ \Omega$. This motor is controlled by a chopper with an input voltage of 230V and frequency of 500Hz. Assuming continuous conduction throughout, Calculate the plot speed torque characteristics for :
i) Motoring operation at duty ratio 0.3
ii) Regenerative braking operation at duty ratio of 0.7.
3. a) What is meant by soft start? State and explain the soft start schemes employed for induction motor.
b) Draw the power circuit of a three-phase current source inverter and explain its working. State also its inherent advantages over voltage source inverter for speed control of induction motor.
4. Explain how regenerative braking is done in three phase induction motor.
5. Describe the load commuted CSI fed synchronous motor drive with neat circuit diagram and explain its working operation with relevant waveform and speed to torque characteristics.
6. Explain working of single phase semi controlled converter connected to d.c. separately excited motor.
7. Explain VSI fed self controlled synchronous motor drive with neat circuit diagram.
OR
Discuss the significance of electric braking and plugging.
8. Describe stator voltage control technique using AC voltage controller for the speed control of a 3- Φ induction motor.
